



OBJECT ORIENTED SOFTWARE ENGINEERING

Ms. Archana A

Department of Computer Applications

OBJECT ORIENTED SOFTWARE ENGINEERING

Incremental Process Model

Archana A

Department of Computer Applications

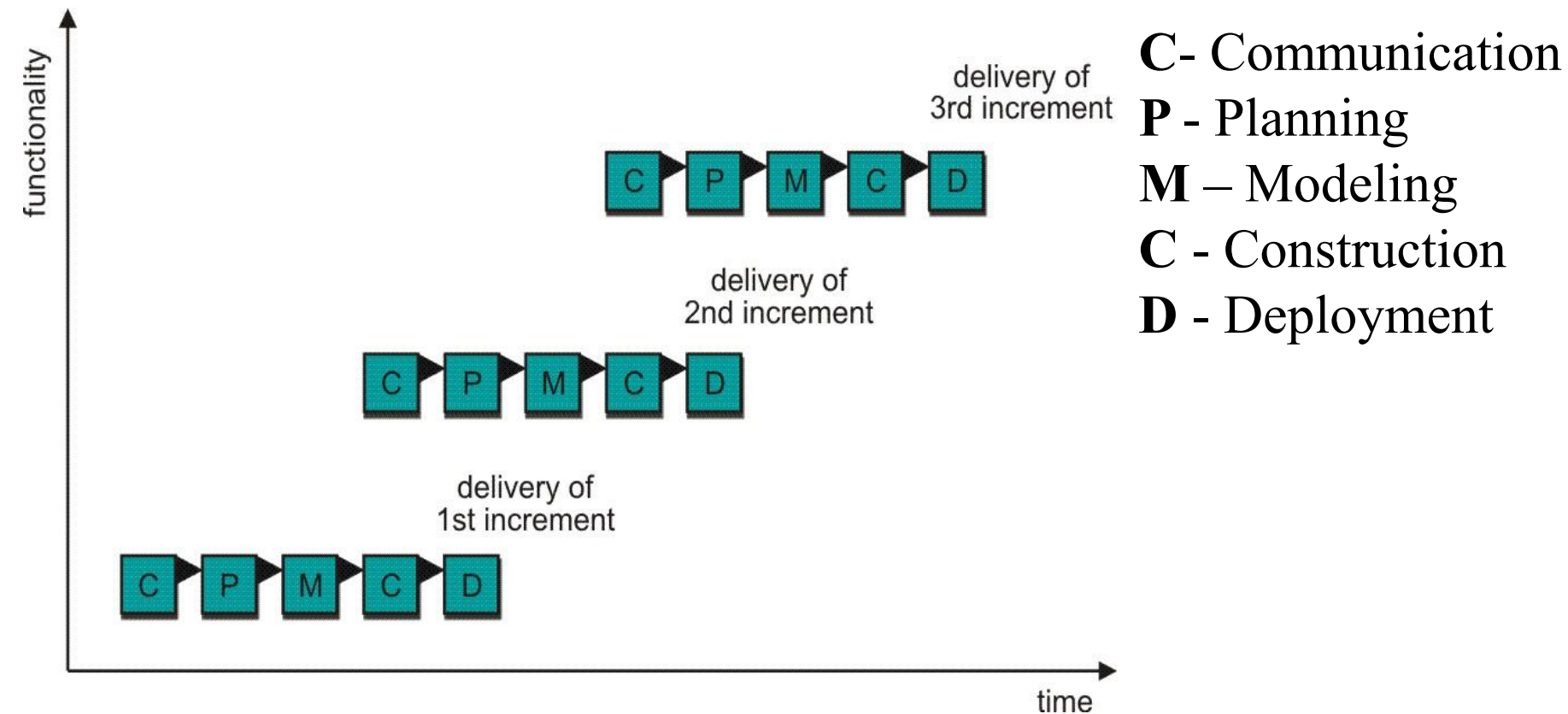
- The incremental model delivers a series of releases, called **increments**.
- Rather than deliver the system as a **single delivery**, the development and delivery is **broken down into increments** with each increment delivering part of the required functionality

- **First Increment** is often **core product**:
 - Includes **basic requirement**,
 - **Many supplementary features** (known & unknown) remain undelivered.
- A plan of **next increment** is prepared
 - Modifications** of the first increment,
 - Additional features** of the first increment

- It is particularly useful when enough staffing is not available for the whole project.
- Increment can be planned to **manage technical risks**.
- Incremental model focus more on delivery of **operation product** with each increment.

OBJECT ORIENTED SOFTWARE ENGINEERING

3. Incremental Model



Delivers software in **small but usable pieces**, each piece builds on pieces already delivered

For example: word-processing software developed → **4 increments**

- 1st increment → Deliver basic file management, editing, and document production functions.
- 2nd increment → More sophisticated editing and document production capabilities.
- 3rd increment → Spelling and grammar checking.
- 4th increment → Advanced page layout capability.

- Using this model some **working functionality** can be developed **quickly** and **early** in the life cycle.
- **Results are obtained early and periodically.** Parallel development can be planned.
- Progress can be **measured**.
- **Risks are identified** and **resolved** during **iteration**. Easier to manage risk - High risk part is done first. i.e., **Risk analysis is better.**
- It is **suited for large and mission-critical projects**

- In this model more **resources and management attention** is required.
- System architecture or **design issues** may arise because **not all requirements are gathered** in the beginning of the entire life cycle.
- Defining increments may require definition of the complete system.
- It is **not suitable for smaller projects**.
- **Highly skilled resources** are required for **risk analysis**.



THANK YOU

Ms. Archana A

Department of Computer Applications

archana@pes.edu

+91 80 6666 3333 Extn 392